Aging as a Decoding Threshold

A unified theory of repair scheduling in self-maintaining networks, joining coding theory, percolation, tropical algebra, large deviations, ultrametric analysis, and reinforcement learning

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Abstract. Aging is usually modelled as damage that outpaces repair. We ask a sharper question: at a fixed total repair capacity, does the *order* in which damage is repaired change lifespan, by how much, and why? We model a self-maintaining system as units whose state is protected by redundancy (an erasure code), coupled through a graph along which damage spreads, and serviced by a shared repair machinery of finite throughput. Six mathematical fields enter one framework: coding theory gives a decoding threshold ε^*, percolation a cascade threshold p_c, tropical (max-plus) algebra the throughput efficiency η of a repair cycle, large-deviation theory (a min-plus calculus) the lifespan exponent Φ, ultrametric analysis the spectrum of lineage-structured damage, and stochastic control with reinforcement learning the optimal repair order. We prove that ordering is irrelevant for exchangeable units (so any gain measures structure), derive a closed-form Haar spectrum for ultrametric contagion, an equal-exponent principle for splitting capacity, and the bound $\eta \le 1$, and verify each numerically. In a stylised model, only 21.7% of 5,040 priority orders of seven repair tasks are stall-free, and the worst loses 36.7% of throughput at equal total work. Exact dynamic programming lengthens the expected lifetime of a two-class system by 78% over the best static split. A policy learned by cross-entropy search lengthens the mean lifetime of a 64-unit hierarchical network by 17–18% over random order, while intuitive "organ-first" heuristics shorten it by 6–9%. The gain falls with contagion strength and rises with repair budget, contrary to our initial expectation. A composite threshold conjecture, $\theta_{\text{eff}} = \min(\varepsilon^*, p_c)$, is verified only component-wise. We provide a pre-registrable *C. elegans* protocol with a power analysis. All results are in silico; no claim is made about human lifespan.

Keywords: aging · repair scheduling · erasure codes · percolation · max-plus algebra · large deviations · ultrametric spaces · dynamic programming · reinforcement learning

+78%	+17–18%	37%	22%
exact optimal vs. best static split (two-class model)	learned policy vs. random order (64-unit network)	throughput lost by the worst of 5,040 orders, equal work	of priority orders are stall-free

Scope. *This is a theoretical and computational preprint. Every number reported is produced by the models defined here, with dimensionless, stylised parameters; none comes from an experiment on living organisms. The framework yields falsifiable predictions (Section 13) but does not by itself show that any biological repair order can be changed to extend life.*

Contents

1 Introduction . . . 3

2 Related work and positioning . . . 4

2.1 What is new, and what is not . . . 4

3 A model of self-maintaining systems . . . 5

3.1 Units, damage and repair . . . 5

3.2 Redundancy and failure . . . 5

3.3 The composite conjecture . . . 5

3.4 When ordering cannot matter . . . 5

4 Layer 1: redundancy as an erasure code . . . 6

4.1 Template-assisted repair as peeling decoding . . . 6

4.2 Numerical results . . . 6

5 Layer 2: damage percolation . . . 8

6 Layer 3: throughput limits and large deviations . . . 8

7 Layer 4: tropical algebra and the efficiency of a repair cycle . . . 10

7.1 A timed event graph of shared repair machinery . . . 10

7.2 Exhaustive enumeration . . . 11

8 Layer 5: min-plus lifespans and the equal-exponent principle . . . 12

9 Layer 6: ultrametric lineage structure . . . 13

10 Optimal repair order: dynamic programming and the HJB equation . . . 14

11 Reinforcement learning on a hierarchical network . . . 15

11.1 Robustness: contagion and budget . . . 17

12 The composite formula applied, and age-dependent capacity . . . 18

12.1 Topology, efficiency and the exponent . . . 18

12.2 Gompertz-like mortality from declining capacity . . . 19

13 Predictions, status, and an experimental protocol . . . 20

13.1 A pre-registrable experiment . . . 20

14 Limitations . . . 21

15 Discussion . . . 21

16 Conclusion . . . 22

16.1 Acknowledgments and disclosures . . . 22

References . . . 22

Appendix A: Proofs . . . 24

Appendix B: Algorithms and reproducibility . . . 24

1 Introduction

Aging is commonly described as the progressive accumulation of molecular and cellular damage that outpaces the capacity to repair it [14, 15]. Mathematical treatments have long drawn on reliability engineering and information theory: von Neumann asked how reliable organisms can be built from unreliable parts [7], Szilard and Orgel proposed that errors can feed back and become self-amplifying [8, 9], and the reliability theory of aging explains Gompertz-type mortality [13] from redundancy and failure of components [12]. Interventions that extend life in model organisms, from insulin-signalling mutants [17] to rapamycin [18], are usually interpreted as changing the *amount* of damage or repair.

One question is rarely isolated: **at a fixed total repair capacity, does the order in which damage is repaired matter?** In computing and manufacturing, scheduling is an entire discipline precisely because order matters even when the total work is fixed {c('graham')}. Cellular repair machinery is shared (proteasomes, autophagic capacity and DNA-repair complexes serve many substrates), it obeys precedence constraints (recognition precedes excision, which precedes resynthesis), and the damage it serves is correlated in space and in lineage. These are the ingredients of a scheduling problem.

We therefore pose three sub-questions. *When* does ordering matter at all? *How much* can it matter, and what limits the gain? *Which* mathematical structures determine the answer? To make progress without drowning in biological detail, we build the smallest model that retains four features: redundancy that lets a damaged unit be reconstructed, a network along which damage spreads, a shared repair machinery with finite throughput and precedence, and a failure criterion. Each feature turns out to be the natural home of a different field of mathematics.

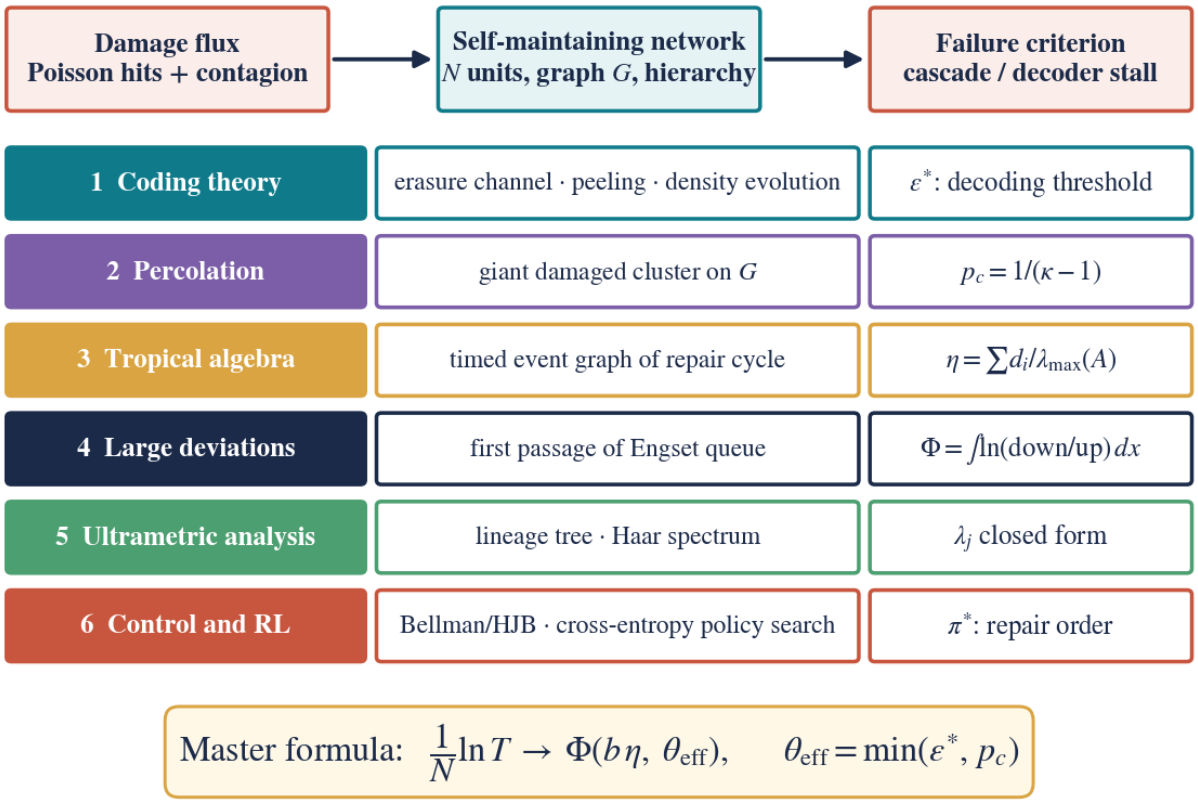


Figure 1. The framework as a stack of six layers. Damage flows into a networked, redundant system. Each layer contributes one quantity to the composite lifespan exponent of Eq. (4). Layers 1 to 4 are quantified analytically and checked numerically; layers 5 and 6 are applied in Sections 9 to 11.

Contributions. (i) A formal model in which lifespan is a first-passage time of a controlled, networked, coded system (Section 3). (ii) Proposition 1: for exchangeable units every work-conserving policy yields the same lifespan law, so any benefit of ordering is a measure of structure. (iii) A closed-form Haar spectrum for contagion

on an ultrametric tree (Proposition 2) and an exact large-deviation law for the throughput channel (Proposition 3). (iv) A tropical (max-plus) efficiency $\eta \leq 1$ for repair cycles, with exhaustive enumeration over all 5,040 orders of a seven-task system (Section 7). (v) An equal-exponent principle for splitting repair capacity between failure channels, a min-plus statement (Proposition 5). (vi) Exact dynamic programming and policy-search reinforcement learning, including negative results (Sections 10 and 11). (vii) A composite threshold conjecture, a list of falsifiable predictions with their current status, and a pre-registrable experimental protocol (Section 13).

How to read the evidence. Statements are labelled *Proposition* (proved in Appendix A), *Conjecture* (not proved), or *Numerical result* (produced by code with the stated seeds and sizes). Where an outcome contradicts our expectation, we report it as such. We did not perform an exhaustive literature search, so novelty is claimed only as "to our knowledge".

2 Related work and positioning

Table 1 lists the established results on which we build, how each is used here, and what we do *not* claim. The closest prior ideas are the reliability theory of aging [12], which also treats redundancy and failure thresholds, the information theory of aging, which interprets aging as loss of epigenetic information [16], and kinetic proofreading [10], a classic example of error-reducing biochemistry that spends energy for accuracy.

Table 1. Positioning against established results.

Established result	Used here as	Not claimed
Reliability theory of aging [12]; error catastrophe [9]	Motivation for a failure threshold and for an unrepairable damage core	That our thresholds are those of real cells
Information loss as a driver of aging [16]	Supports reading damage as loss of information, hence the erasure channel	A specific molecular code or its rate
LDPC codes, density evolution on the erasure channel [2, 3, 4, 6]	Peeling decoder as a model of template-assisted repair; threshold ε^*	That cells implement LDPC codes
Percolation on random graphs [24, 20, 21, 25]; immunization [23]	Cascade threshold $p_c = 1/(\kappa-1)$; repair as dynamic immunization	Validity for clustered or correlated networks
Max-plus algebra and timed event graphs [30, 31, 33]	Cycle time of a repair loop with shared machinery and precedence	That real repair cycles are deterministic
Finite-source queue [36], large deviations [35]	Exact first-passage law of damage load under limited throughput	Exponential tail for human mortality
Restless bandits [40]; dynamic programming [37, 38, 39]	Optimal allocation among damaged classes	Index-policy optimality here
Reinforcement learning, cross-entropy method [42, 43]	Policy search over a feature-based priority rule	Sample efficiency or deployment

2.1 What is new, and what is not

Each ingredient above is standard. Our contribution is their composition around one question, ordering at fixed flux, together with specific results that follow: the ordering-invariance proposition, the tropical efficiency of repair cycles, the equal-exponent principle, and the numerical comparison of exact and learned repair policies. We have not found this composition in the literature we know, but we cannot rule out that parts of it exist in work we did not consult. We also do not claim that any of the stylised numbers below transfer to biology.

3 A model of self-maintaining systems

3.1 Units, damage and repair

The system has N units (molecules, organelles or cells, depending on the scale of interest). Unit i is intact ($s_i = 0$) or damaged ($s_i = 1$). The state evolves as a continuous-time Markov process in which an intact unit becomes damaged at rate λ plus a contagion term, and a damaged unit that the policy π currently serves is repaired at rate μ:

$$s_i:\ 0\to 1\ \text{ at rate }\ \lambda+\beta\sum_j W_{ij}s_j, \qquad s_i:\ 1\to 0\ \text{ at rate }\ \mu\,\mathbf{1}\{i\in\pi(s)\}, \quad |\pi(s)|\le B. \tag{1}$$

The policy $\pi(s)$ selects at most B damaged units (the repair slots). A policy is *work-conserving* if it never leaves a slot idle while a damaged unit is waiting. The contagion kernel W encodes network or lineage structure; in the hierarchical model of Section 9 it depends only on the tree distance:

$$W_{ij}=\frac{2^{-a\,d(i,j)}}{Z_i}, \qquad Z_i=\sum_{k\neq i}2^{-a\,d(i,k)}, \qquad d(i,j)=\min\{\ell:\ \lfloor i/2^{\ell}\rfloor=\lfloor j/2^{\ell}\rfloor\}. \tag{2}$$

3.2 Redundancy and failure

Functional integrity is protected by redundancy: a damaged unit can be reconstructed if enough of its redundant relatives are intact. In Section 4 this is formalised as an erasure code and decoded by a peeling algorithm. The organism fails when the damage state enters a failure set F; the lifespan is the first-passage time

$$T=\inf\{t\ge 0:\ s(t)\in F\}, \qquad F=\{D\ge\theta N\}\ \text{ or }\ \{\text{a module is overloaded}\}, \quad D=\sum_i s_i. \tag{3}$$

3.3 The composite conjecture

The layers of Figure 1 combine into one statement. With budget per unit $b = B/N$ and tropical efficiency η of the repair cycle (Section 7), the effective budget is $b\eta$. The failure fraction is limited both by the decoding threshold ε^* and by the percolation threshold p_c of the damage network. We conjecture

$$\frac{1}{N}\ln \mathbb{E}\,T \longrightarrow \Phi(b\eta,\ \theta_{\text{eff}}), \qquad \theta_{\text{eff}}=\min(\varepsilon^*,\ p_c). \tag{4}$$

> ***Conjecture 1* (composite threshold).** *Eq. (4) holds for large N in the regime where damage is rare relative to repair throughput. The queueing factor Φ is exact (Proposition 3), and the thresholds ε^* and p_c are separately verified (Sections 4 and 5). We have **not** verified the composition end to end with a single simulation that includes all layers, and we regard it as a conjecture.*

3.4 When ordering cannot matter

> ***Proposition 1* (ordering invariance).** *Suppose units are exchangeable, the damage-arrival rates and the failure set depend on the state only through the damaged count D, and the policy is work-conserving and non-anticipating. Then D(t) is a birth–death chain whose law does not depend on the policy, so the law of T is the same for every such policy.*

The proof (Appendix A) is short, and its consequence is important: any measured benefit of an ordering is a measure of *structure* (heterogeneity, hierarchy, correlation, or precedence). This gives a clean null hypothesis for experiments (Section 13) and explains why the sections that follow all introduce structure.

Table 2. Notation used throughout. Rates are in units of the single-slot repair rate $\mu = 1$.

Symbol	Meaning	Symbol	Meaning
N	number of repairable units	ε^*	decoding (erasure) threshold of the redundancy code
$s_i \in \{0,1\}$	damage indicator of unit i	p_c, κ	site-percolation threshold, $\kappa = \langle k^2\rangle/\langle k\rangle$
λ, β, μ	intrinsic damage, contagion, repair rates	θ_{eff}	effective failure fraction, $\min(\varepsilon^*, p_c)$
B, $b = B/N$	repair slots; budget per unit	A, $\lambda(A)$	max-plus matrix; its cycle time (eigenvalue)
W_{ij}	contagion kernel (row-normalised)	η	tropical efficiency $\sum d_i / \lambda(A) \le 1$
$d(i,j)$	ultrametric (tree) distance	Φ	lifespan exponent, $(1/N)\ln \mathrm{E}T \to \Phi$
π	repair policy (priority rule)	c_k	exponent of failure channel k
T	lifespan (first-passage time)	α	share of budget given to class 1

4 Layer 1: redundancy as an erasure code

4.1 Template-assisted repair as peeling decoding

Many repair mechanisms restore a lesion by consulting a redundant source: excision repair of a damaged DNA strand uses the complementary strand, homologous recombination uses a second copy, and protein turnover resynthesises from a template. If damage *erases* information at a unit and a repair step can recover one erased unit whenever a redundancy relation (a "check") involves exactly one erased unit, repair is precisely the peeling decoder of a low-density parity-check code on the binary erasure channel {c('gallager','luby','ru01')}. This is an analogy, offered because it yields a sharp, computable threshold; we do not claim that cells implement such codes.

For a regular ensemble with variable degree d_v and check degree d_c, an erasure fraction ε is fully repaired with high probability for large N if and only if ε lies below the density-evolution threshold. The message erasure probability obeys

$$x_{\ell+1} = \varepsilon\left[1 - (1 - x_\ell)^{d_c - 1}\right]^{d_v - 1}, \qquad \varepsilon^* = \sup\{\varepsilon : x_\ell \to 0\}. \tag{5}$$

> ***Proposition 6* (Shannon bound on tolerable damage).** *For any redundancy scheme of design rate R, vanishing repair failure requires $\varepsilon \quad 1 - R$. For a regular (d_v, d_c) ensemble, $\varepsilon^* \quad d_v/d_c$.*

This is the channel-capacity bound of the erasure channel [1] applied to repair; the proof is recalled in Appendix A. It states a universal trade-off: the more of a unit's capacity is devoted to function (higher rate R), the smaller the damage fraction it can tolerate. Redundancy is not free.

4.2 Numerical results

We computed thresholds by bisection on Eq. (5) and checked them with a synchronous peeling decoder on random (3,6) graphs of $N = 500$, 2000 and 8000 units (300, 150 and 60 trials per damage fraction). For the (3,6) ensemble the threshold is $\varepsilon^* = 0.4294$, and the empirical repair-success curves of Figure 2a sharpen around it as N grows. The number of peeling rounds rises steeply approaching ε^* (critical slowing down, panel b), and above ε^* a linear fraction of units remains unrepaired, matching the density-evolution prediction (panel c). This residual set is the analogue of an error catastrophe [9]: damage that no further repair step can remove.

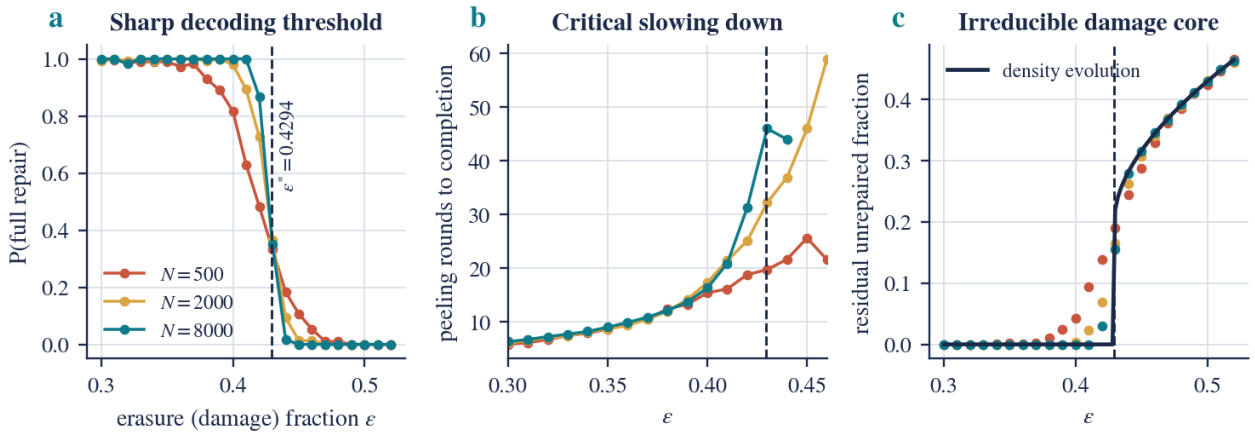


Figure 2. Decoding threshold of the (3,6) ensemble. (a) Probability that all damage is repaired; curves for $N = 500, 2000, 8000$ cross near $\varepsilon^* = 0.4294$ (dashed). (b) Mean number of synchronous peeling rounds among successful decodings. (c) Residual unrepaired fraction: simulation (points) against the density-evolution fixed point (line).

Table 3. Erasure thresholds of regular ensembles from density evolution, with the Shannon limit $1-R$. Lower-rate ensembles tolerate more damage, at the cost of more redundancy per unit of function.

(d_v, d_c)	Design rate R	Threshold ε^*	Shannon limit $1-R$	Gap
(2, 4)	0.500	0.3332	0.500	0.1668
(3, 4)	0.250	0.6474	0.750	0.1026
(3, 5)	0.400	0.5176	0.600	0.0824
(3, 6)	0.500	0.4294	0.500	0.0706
(4, 6)	0.333	0.5061	0.667	0.1605
(4, 8)	0.500	0.3834	0.500	0.1166
(5, 10)	0.500	0.3416	0.500	0.1584

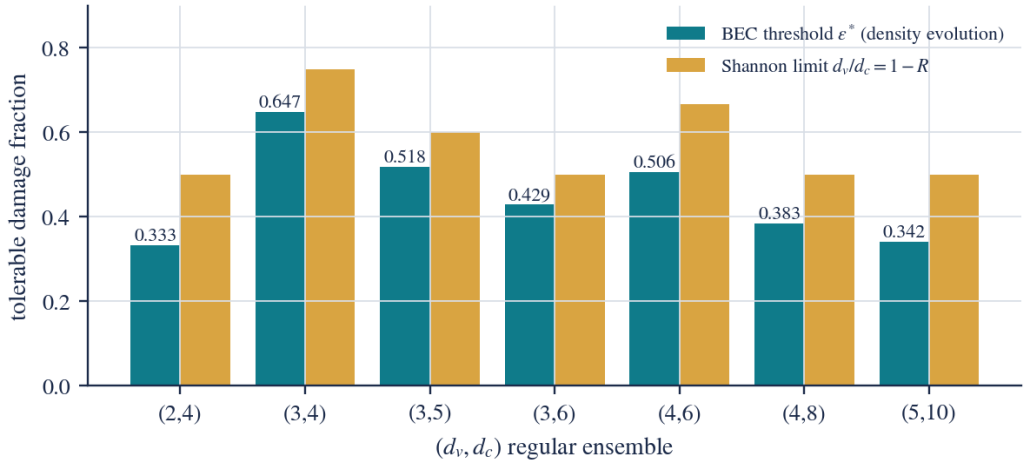


Figure 3. Thresholds against capacity. Bars compare the density-evolution threshold with the Shannon limit for seven regular ensembles. No ensemble exceeds the limit, in agreement with Proposition 6.

The practical message for the rest of the paper is that the coding layer fixes a failure fraction ε^* that is *derived* from a redundancy architecture rather than assumed.

5 Layer 2: damage percolation

When damage spreads along edges, failure need not require a large damaged *fraction*: it suffices that damaged units form a spanning cluster. For site percolation on a random graph with degree moments $\langle k \rangle$ and $\langle k^2 \rangle$, a giant damaged cluster appears at damaged density [24, 20, 21]

$$p_c = \frac{1}{\kappa - 1}, \qquad \kappa = \frac{\langle k^2 \rangle}{\langle k \rangle}. \tag{6}$$

Hubs raise κ and therefore lower p_c, which is why scale-free networks are fragile to random damage [22]. Repair that removes damage preferentially at hubs acts as dynamic immunization [23]. We tested Eq. (6) on four graphs of $N = 20{,}000$ units: Erdős–Rényi [27], random 4-regular, Barabási–Albert [29] and Watts–Strogatz [28], with the measured threshold taken as the density at which the second-largest cluster peaks (grid step 0.02, five replicates).

Table 4. Percolation thresholds: tree-like prediction against simulation.

Topology	κ	p_c predicted	p_c measured	Agreement
Erdős–Rényi (c=4)	4.98	0.251	0.26	good
Random 4-regular	4.00	0.333	0.34	good
Barabási–Albert (m=2)	13.87	0.078	0.18	poor
Watts–Strogatz (k=4, p=0.1)	4.09	0.323	0.52	poor

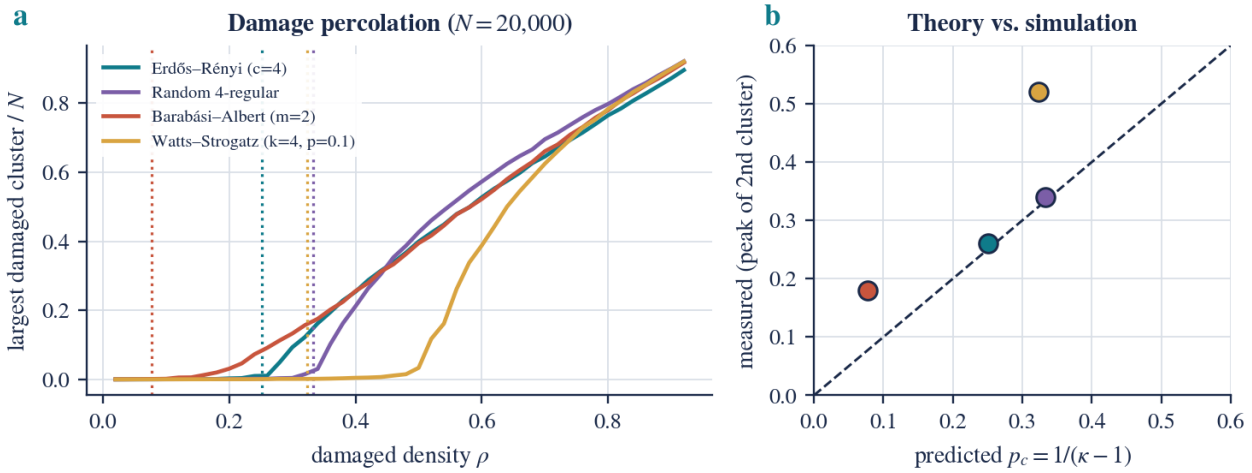


Figure 4. Damage percolation. (a) Largest damaged cluster against damaged density; dotted lines are the predicted thresholds. (b) Predicted against measured threshold; the dashed line is perfect agreement.

Numerical result. The prediction is accurate for the Erdős–Rényi and random-regular graphs, whose structure is locally tree-like. It fails for the Barabási–Albert graph (predicted 0.078, measured about 0.18), where finite size and degree correlations matter, and for the Watts–Strogatz graph (predicted 0.32, measured about 0.52), where clustering violates the tree approximation. For the clustered graph the formula is conservative, since it predicts failure at lower damage than observed. Any use of Eq. (6) for a real network should therefore be preceded by a direct percolation measurement.

6 Layer 3: throughput limits and large deviations

Consider the damaged count D of N exchangeable units with B repair slots. It is a birth–death chain (a finite-source queue [36]) with up-rate and down-rate

$$u_D = \lambda (N - D), \qquad d_D = \mu \min(D, B). \tag{7}$$

Failure occurs when D first reaches $m = \lceil \theta N \rceil$. The mean first-passage time is available in closed form, and its large-N limit has a simple rate function.

> ***Proposition 3* (exact lifespan and large-deviation limit).** *Let π_D $k D$ u_{k-1}/d_k. Then ET = $i<m$ $(u_i \pi_i)^{-1}$ $j\, i\, \pi_j$. With x = D/N and b = B/N, if the fluid fixed point x* lies below θ, then (1/N) ln ET Φ = x^* θ ln[μ min(x,b) / (λ(1−x))] dx. If x* θ, then ET grows only logarithmically in N.*

$$\Phi(b,\theta) = \int_{x^*}^{\theta} \ln\left[\frac{\mu \min(x,b)}{\lambda(1-x)}\right] dx. \tag{8}$$

We used $\lambda = 0.25$, $\mu = 1$ and $\theta = 0.4$. At $b = 0.3$ the limit is $\Phi = 0.0896$. The exact finite-size values converge to it (Table 5, Figure 5), and the stationary law agrees with Gillespie simulation [51].

Table 5. Convergence of the lifespan exponent ($\lambda = 0.25$, $\mu = 1$, $\theta = 0.4$, $b = 0.3$).

N	(1/N) ln ET (exact)	Φ (limit)
20	0.1167	0.0896
40	0.0952	0.0896
80	0.0873	0.0896
160	0.0861	0.0896
320	0.0867	0.0896
640	0.0876	0.0896
1280	0.0883	0.0896

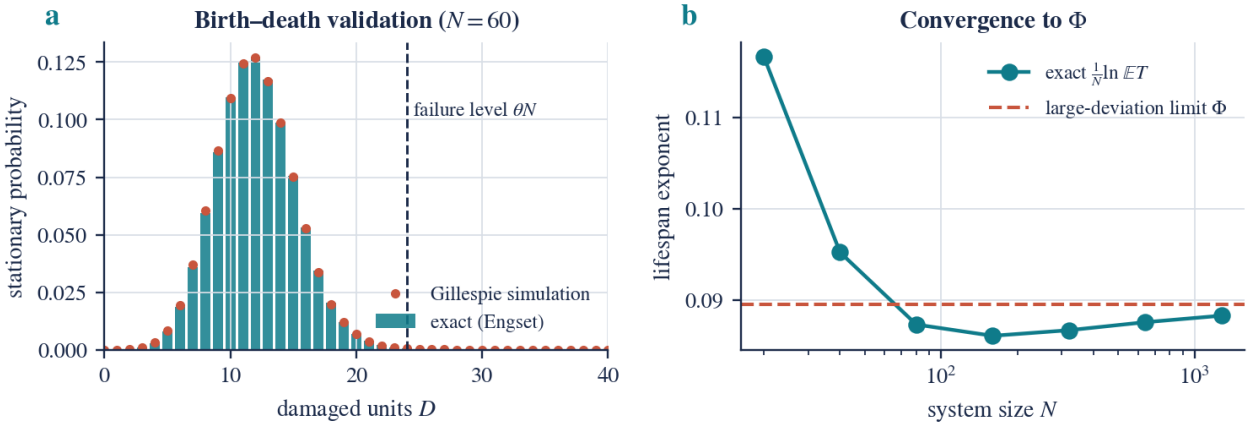


Figure 5. Validation of the throughput layer. (a) Exact stationary distribution (bars) and Gillespie simulation (points) for $N = 60$. (b) Exact lifespan exponent against system size, converging to the large-deviation limit.

Figure 6 shows the exponent as a function of budget and exposes three regimes. Below $b = \lambda(1-\theta)/\mu = 0.15$ the system is in **overload**: the fluid state already lies beyond the failure level and lifespan is short. Between 0.15 and $\theta = 0.4$, each extra unit of budget buys an **exponential** gain in lifespan. Above $b = \theta$ the exponent **saturates**: additional capacity cannot repair more than the damaged units that exist. Budget alone therefore has diminishing returns, which motivates studying how the budget is *used*.

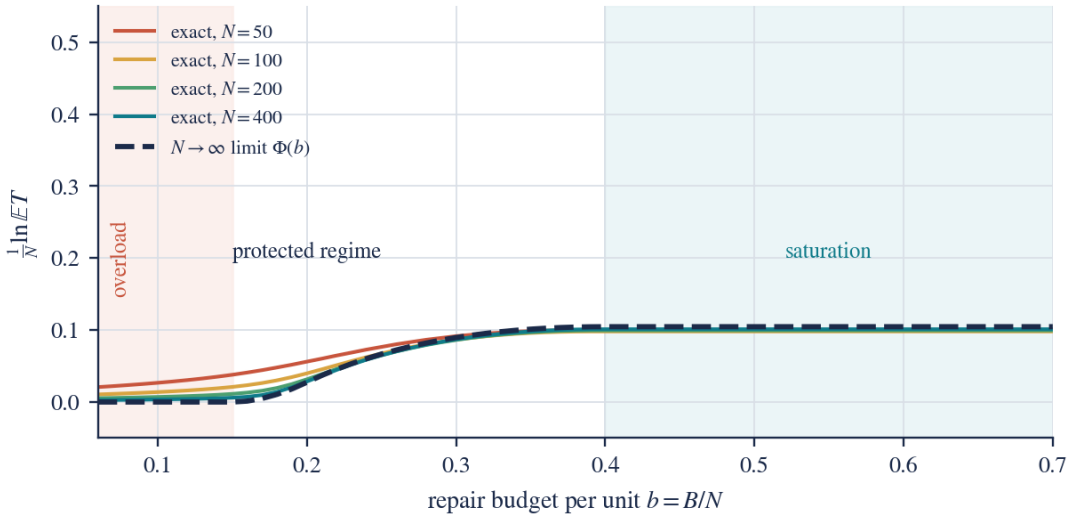


Figure 6. Lifespan exponent against repair budget. Exact curves for four system sizes approach the dashed large-deviation limit. Three regimes are shaded: overload, protected, and saturation.

7 Layer 4: tropical algebra and the efficiency of a repair cycle

7.1 A timed event graph of shared repair machinery

Real repair is a cycle of tasks that share machinery and obey precedence with delays. We model seven tasks (Table 6) executed in a cyclic priority order σ by one shared machine, so each task starts only after its predecessor in σ has finished. In addition, biological precedence arcs $i \to j$ require task j to wait a lag after task i finishes; lags consume no machine time. An arc with token 1 refers to the previous cycle. The start times $x(k)$ in cycle k obey a max-plus linear system [30, 31]:

$$x(k) = A_0 \otimes x(k) \oplus A_1 \otimes x(k-1) \implies x(k) = A \otimes x(k-1), \qquad A = A_0^{*} \otimes A_1, \tag{9}$$

where $\oplus$ is maximum, $\otimes$ is addition, and A_0^{*} is the Kleene star (longest-path closure) of the zero-token arcs. The parameters in Table 6 are stylised and dimensionless; they illustrate the algebra and are not measurements.

Table 6. The seven stylised repair tasks, durations and precedence arcs. The arcs listed on a row start at that row's task, except where two arcs are shown.

Task	Abbrev.	Duration d	Precedence arc (lag)
Chaperone refolding	CR	2	CR → UP (1.5)
Proteasomal clearance	UP	3	UP → AI (2.0)
Autophagy initiation	AI	1.5	AI → AM (3.0)
Autophagosome maturation	AM	2.5	MF → AM (1.0); AM → CR (1.0)
Mitochondrial fission	MF	1	–
Base-excision repair	BE	2	BE → DB (2.5)
DSB repair	DB	3.5	DB → AI (0.5)

***Proposition 4* (cycle time and efficiency).** *For the system (9) the asymptotic cycle time equals the maximum circuit ratio, $\lambda(A) = \max_{k\,n} \max_i (A^k)_{ii}/k$, and the efficiency $\eta = d_i/\lambda(A)$ satisfies $\eta \quad 1$, with equality if and only if the order σ produces no stall on the critical circuit.*

$$\lambda(A) = \max_{1 \le k \le n} \max_i \frac{(A^{\otimes k})_{ii}}{k}, \qquad \eta(\sigma) = \frac{\sum_i d_i}{\lambda(A_\sigma)} \le 1. \tag{10}$$

7.2 Exhaustive enumeration

We enumerated all 5,040 priority orders. Total work is 15.5 time units for every order, yet the cycle time ranges from 15.5 (η = 1) to 24.5 (η = 0.633). The max-plus eigenvalue agreed with an independent parametric Bellman–Ford computation to within 1e-12 on 150 randomly chosen orders. A fraction 21.7% of orders (1092) are stall-free, the median efficiency is 0.886, and the order in which the tasks are listed in Table 6 is the worst. The reason is visible in Figure 7: in the worst order the critical circuit passes through several lags while the machine idles, whereas the best order fills those lags with independent work.

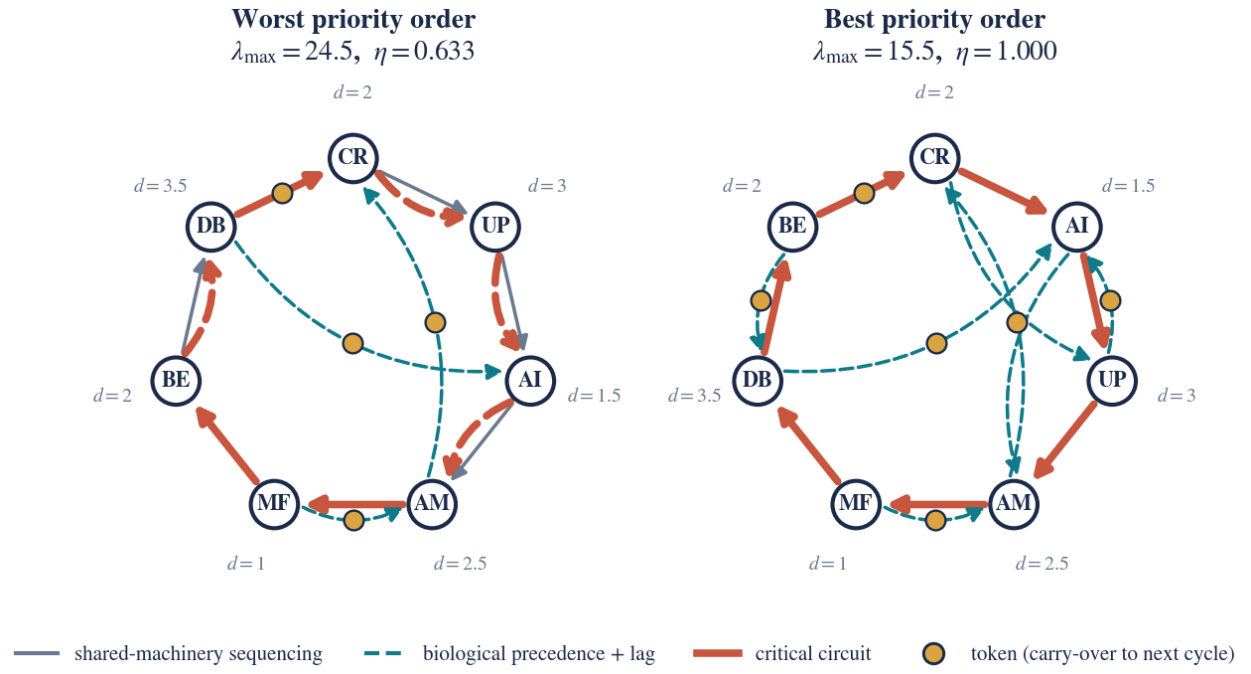


Figure 7. Timed event graphs for the worst and best priority orders. Grey arcs sequence the shared machinery; dashed teal arcs are biological precedence with lag; gold dots mark tokens; the critical circuit is red.

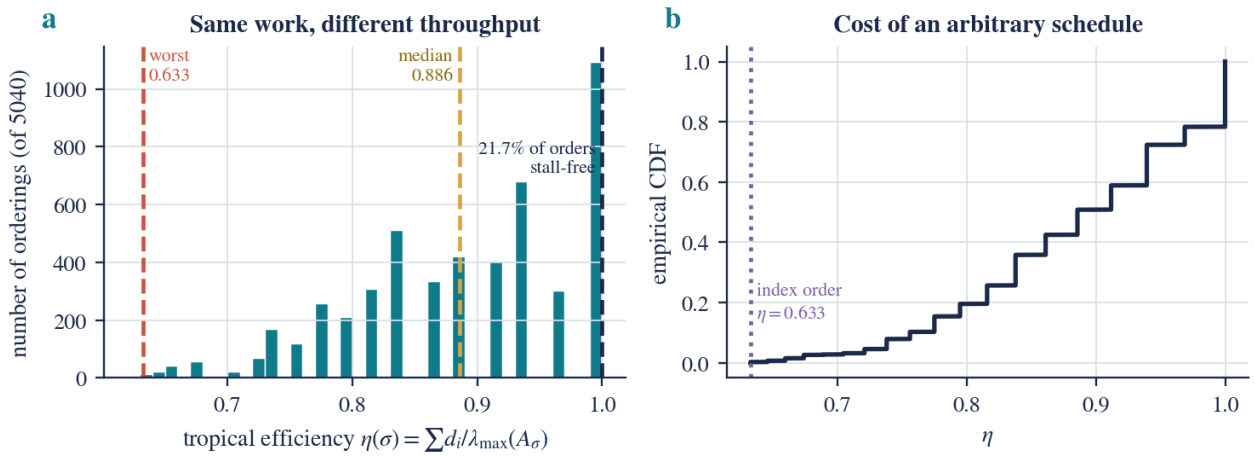


Figure 8. Efficiency across all 5,040 orders. (a) Histogram of η; (b) empirical distribution function, with the index order marked.

Table 7. Summary of the tropical enumeration.

Quantity	Value
Total work $\sum d$	15.5
Best cycle time ($\eta = 1$)	15.5
Worst cycle time	24.5 ($\eta = 0.633$)
Median η	0.886
Stall-free orders	1092 of 5040 (21.7%)
Best order	CR, AI, UP, AM, MF, DB, BE
Max-plus vs. Bellman–Ford error	1.0e-12

In the composite formula, η rescales the budget: a schedule with $\eta = 0.633$ behaves, for throughput purposes, like a budget reduced by 37%, which Section 12 translates into lifespan exponents.

8 Layer 5: min-plus lifespans and the equal-exponent principle

A system with two independent failure channels, such as a critical class and a peripheral class, fails when the first channel fails. If channel k has lifetime exponent c_k, the hazards add, so $\mathrm{E}T \approx (\sum_k 1/\mathrm{E}T_k)^{-1}$. In logarithmic scale this is a min-plus (tropical) sum, via Maslov dequantization [34, 32]:

$$\frac{1}{N}\ln \mathbb{E}\,T \to \min_k c_k\,, \qquad -\varepsilon \ln \sum_k e^{-c_k/\varepsilon} \to \min_k c_k \quad (\varepsilon \to 0). \tag{11}$$

> ***Proposition 5*** ***(equal-exponent principle).*** *Let the budget b be split as αb and $(1-\alpha)b$ between two channels, and let $c_1(\alpha b)$ be strictly increasing and $c_2((1-\alpha)b)$ strictly decreasing in α on the unsaturated region. Then the static split maximizing $\min(c_1, c_2)$ is the unique α^* with $c_1(\alpha^* b) = c_2((1-\alpha^*)b)$.*

Table 8. Lifespan exponents for static budget splits between two failure channels (shares 0.4 and 0.6; $\lambda = 0.20, 0.15$; $\theta = 0.5, 0.7$; $\mu = 1$). The multiplier is $\exp(N\Delta)$, an asymptotic statement.

b	α^*	Exponent at α^*	$\alpha = 0.4$	$\alpha = 0.5$	$\alpha = 0.8$	Multiplier vs. even split ($N = 100$)
0.13	0.60	0.0492	0.0075	0.0279	0.0000	8.4×
0.18	0.64	0.0920	0.0412	0.0679	0.0082	11.2×
0.23	0.70	0.1126	0.0703	0.0919	0.0308	7.9×
0.28	0.75	0.1176	0.0896	0.1062	0.0620	3.1×

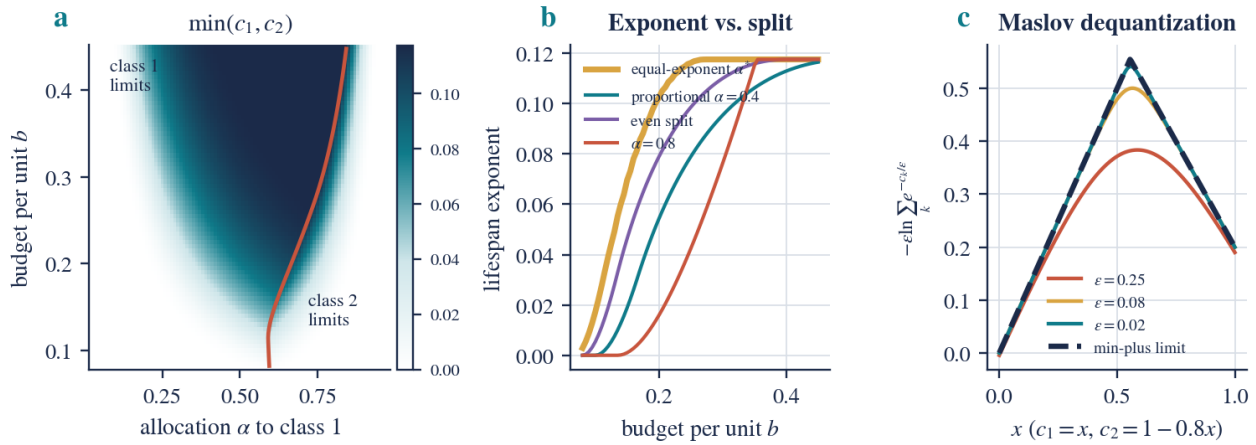


Figure 9. Min-plus structure of the lifespan. (a) Exponent $\min(c_1,c_2)$ over allocation α and budget b; the red curve is the equal-exponent split. (b) Exponent against budget for four static policies. (c) Maslov dequantization: smooth log-sum-exp curves converge to the piecewise-linear minimum.

Numerical result. The optimal split moves toward the more fragile channel as budget grows, and at modest budgets the equal-exponent split beats an even split by an exponent difference that translates into a large multiplier at realistic N. The limitation is that a static split cannot react to the state; Section 10 removes that restriction.

9 Layer 6: ultrametric lineage structure

Damage lineages (molecule, complex, organelle, cell, tissue) form a hierarchy. We model the leaves of a balanced p-ary tree of depth L with ultrametric distance d, so that $d(x,z) \le \max(d(x,y), d(y,z))$ [44, 46, 45]. Contagion with rate w_k per target at distance k has generator

$$(\mathcal{L}f)(x) = \sum_{y \neq x} w_{d(x,y)} \, [f(y) - f(x)]. \tag{12}$$

> ***Proposition 2* (Haar spectrum).** *The eigenfunctions of (12) are the Haar-type wavelets. A wavelet living at level j (constant on blocks of size p^{j-1}, with zero mean over the p sub-blocks of a block of size p^j) has eigenvalue $\lambda_j = -[w_j\, p^j + (p-1) \sum_{k>j} w_k\, p^{k-1}]$, with multiplicity $(p-1)p^{L-j}$. The constant function has eigenvalue 0.*

$$\lambda_j = -\left[w_j p^j + (p-1) \sum_{k>j} w_k p^{k-1} \right], \qquad m_j = (p-1)\, p^{L-j}. \tag{13}$$

The derivation is in Appendix A. We checked Eq. (13) on a binary tree with $L = 5$ (32 leaves) and $w_k = 2^{-1.5k}$: all 32 numerical eigenvalues match the closed form with maximum absolute error 2.4e-15 (Figure 10b). Because the rates λ_j spread over several orders of magnitude, damage perturbations relax on many timescales, and the mean relaxation function decays more slowly than any single exponential (panel c). This multi-timescale relaxation is the reason that block-level information, not only nearest-neighbour information, can matter to a repair policy.

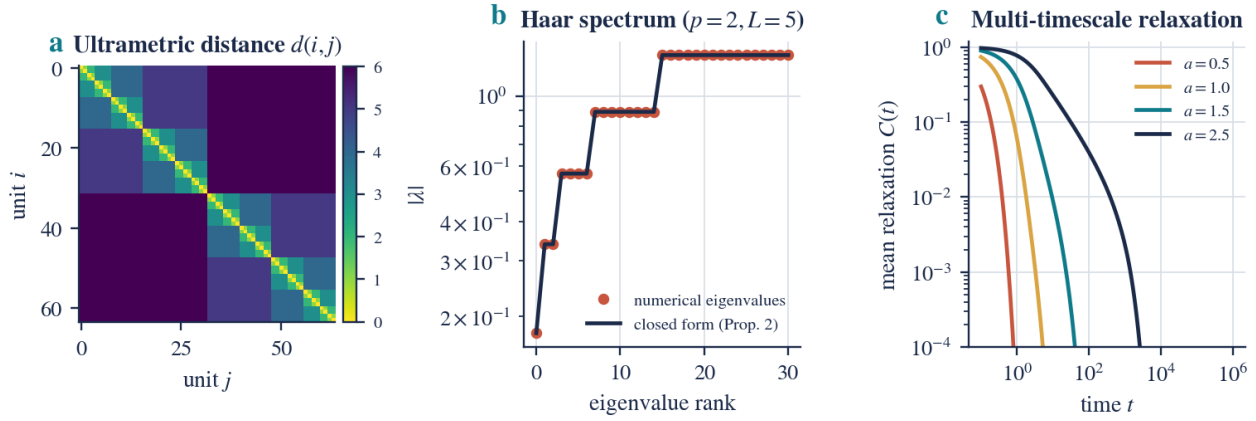


Figure 10. Ultrametric contagion. (a) Distance matrix of 64 leaves of a binary tree, showing nested block structure. (b) Numerical eigenvalues (points) against the closed form of Proposition 2 (line), $p = 2$, $L = 5$. (c) Mean relaxation $C(t)$ for kernel exponents $a =$ 0.5 to 2.5, $L = 6$.

10 Optimal repair order: dynamic programming and the HJB equation

We now let the allocation depend on the state. Consider two classes of units: a critical, upstream class ($N_1 = 25$, failure at $d_1 = 9$) and a peripheral class ($N_2 = 50$, failure at $d_2 = 20$). Damage arrives at rate $\lambda_1 = 0.08$ per intact critical unit and $\lambda_2(1 + \gamma d_1/N_1) = 0.10(1 + d_1/N_1)$ per intact peripheral unit, so critical damage accelerates peripheral damage (a cascade). A controller assigns a share α of the $B = 8$ slots to class 1 and the rest to class 2, with idle slots reassigned (work-conserving). With 180 transient states the expected lifetime under any policy, and the optimal policy, can be computed *exactly* by policy iteration on the Bellman equation [37]:

$$0 = 1 + \max_{\alpha \in [0,1]} \sum_{s'} q(s \to s'; \alpha)\big[V(s') - V(s)\big], \qquad V = 0 \text{ on } F. \tag{14}$$

In the scaling limit ($N \to \infty$ with diffusive fluctuations of order $1/\sqrt{N}$) this becomes the Hamilton–Jacobi–Bellman equation on the unit square [39, 38],

$$0 = 1 + \sup_{\alpha} \Big\{ b(x,\alpha)\cdot\nabla V + \tfrac{1}{2N}\,\mathrm{tr}\big[\Sigma(x,\alpha)\nabla^2 V\big] \Big\}, \qquad V|_{\partial\Omega} = 0, \tag{15}$$

whose discrete counterpart we solve here (the Markov-chain approximation). We did not solve the continuum PDE separately.

Table 9. Exact expected lifetimes for the two-class cascade model. The best static split is $\alpha = 1$ (critical class first).

Policy	Expected lifetime	vs. proportional	vs. best static
Proportional (uniform over damaged units)	209.6	1.00×	0.60×
Static split $\alpha = 0.5$	275.6	1.31×	0.79×
Class-2 first	38.9	0.19×	0.11×
Class-1 (upstream) first	350.0	1.67×	1.00×
Best static split	350.0	1.67×	1.00×
DP-optimal (Bellman)	623.2	2.97×	1.78×

Numerical result. Policy iteration converged in 3 iterations (expected lifetime 276, 617, 623). Proportional repair, which is equivalent to choosing damaged units uniformly at random, gives 210. The best static rule, always serving the critical class first, gives 350. The Bellman-optimal policy gives 623: 2.97 times proportional and 78% above the best static split. The optimal share (Figure 11b) is a state-dependent switching rule that gives full priority to the critical class when it is most threatened and shifts capacity to the peripheral class when that class approaches its own failure level.

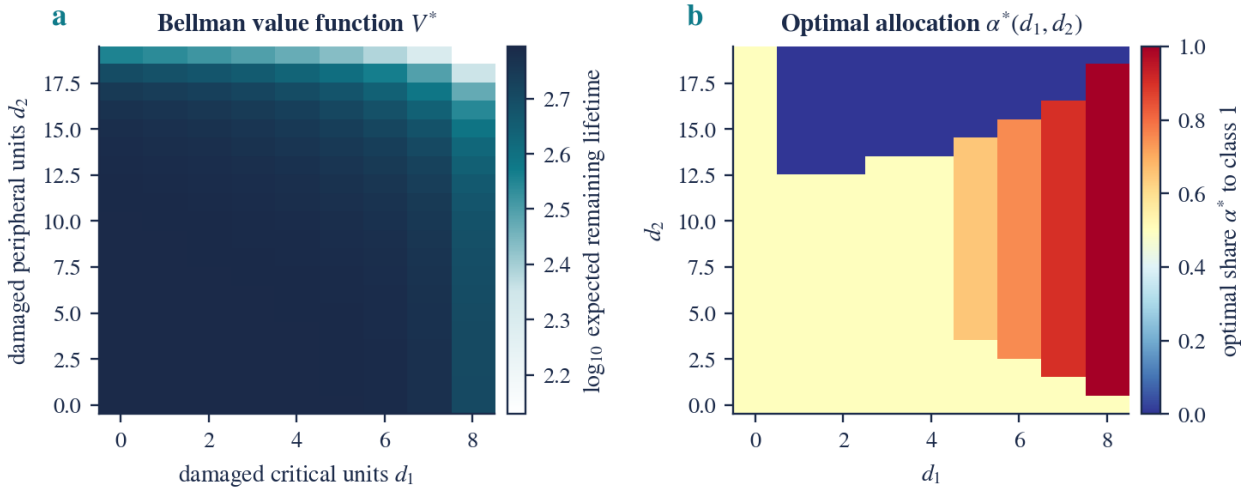


Figure 11. Exact dynamic programming. (a) Logarithm of the expected remaining lifetime under the optimal policy. (b) Optimal share of the budget given to the critical class as a function of the damage state.

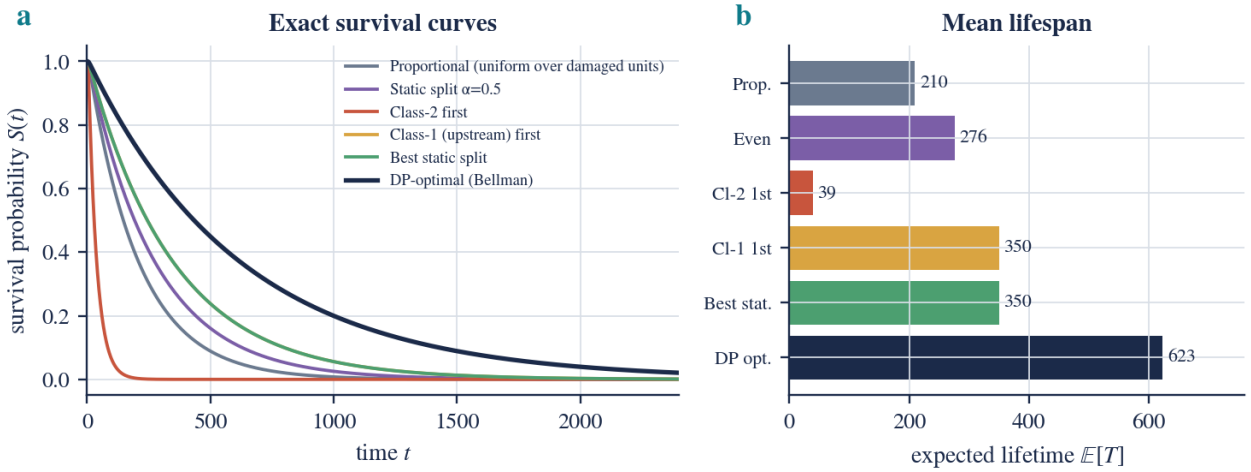


Figure 12. Survival under five repair policies. (a) Exact survival functions from the matrix exponential of the generator. (b) Expected lifetimes.

11 Reinforcement learning on a hierarchical network

The exact solution is possible only for tiny state spaces. For a 64-unit system with ultrametric contagion we use policy search. The system is the model of Section 3 with the kernel of Eq. (2), $a = 1.5$, simulated exactly by the Gillespie method [51] (parameters in Table 10). Failure is the first time any 8-unit block has 7 damaged units or 32 units are damaged. A policy scores each damaged unit by a linear function of six features (the damaged fraction in its blocks of 2, 4, 8 and 16 units; the age of the damage; and its *exposure*, the kernel weight onto still-intact units), and serves the B highest-scoring units. The weight vector is trained by the cross-entropy method [43], a population-based reinforcement-learning algorithm [42] that maximizes mean lifetime.

Table 10. Simulation parameters (dimensionless).

Parameter	Value
Units, hierarchy	$N = 64$, binary tree of depth 6
Kernel	$a = 1.5$, rows normalised to 1
Rates	$\lambda = 0.008$, $\beta = 0.7$, $\mu = 1$
Repair slots	$B = 3$
Failure	8-unit block with ≥ 7 damaged, or ≥ 32 damaged
CEM	50 generations, population 40, elite 8, 300 lives per candidate, fresh seeds each generation
Evaluation	4000 lives per policy on seeds disjoint from training; bootstrap 95% intervals; log-rank test
Independent runs	3 (different random seeds)

Table 11. Lifetimes of repair policies in the 64-unit hierarchical network (4000 lives each).

Policy	Mean lifetime (95% CI)	Median	Ratio to random (95% CI)	Log-rank *p*
Uniform random	40.1 (39.1–41.1)	30.5	–	–
FIFO (oldest first)	39.7 (38.7–40.8)	29.8	0.990 (0.955–1.026)	0.60
LIFO (newest first)	40.5 (39.5–41.5)	30.7	1.010 (0.974–1.046)	0.60
Exposure-greedy	45.9 (44.7–47.2)	33.9	1.144 (1.102–1.184)	7e-13
Block-greedy (organ)	37.7 (36.8–38.6)	28.9	0.939 (0.907–0.974)	4e-04
Lineage-sum (ultrametric)	36.6 (35.7–37.6)	27.8	0.912 (0.880–0.946)	4e-07
RL-CEM (run 1)	47.1 (45.9–48.5)	35.0	1.174 (1.135–1.219)	9e-18
RL-CEM (run 2)	47.5 (46.2–48.7)	35.1	1.183 (1.138–1.226)	4e-19
RL-CEM (run 3)	47.5 (46.2–48.7)	35.0	1.183 (1.138–1.226)	4e-19

Numerical result. Three independent training runs reached the same performance, +17.4% to +18.3% over uniform random order (95% intervals 13.5–22.6%). Of the hand-built rules, only exposure-greedy beat random (14.4%). Block-greedy and lineage-sum, which seem natural because they concentrate repair on the most damaged blocks, were *worse* than random (-6.1% and -8.8%), and FIFO and LIFO were indistinguishable from random. We therefore reject our prior expectation that ultrametric-aware greedy rules beat age-based rules.

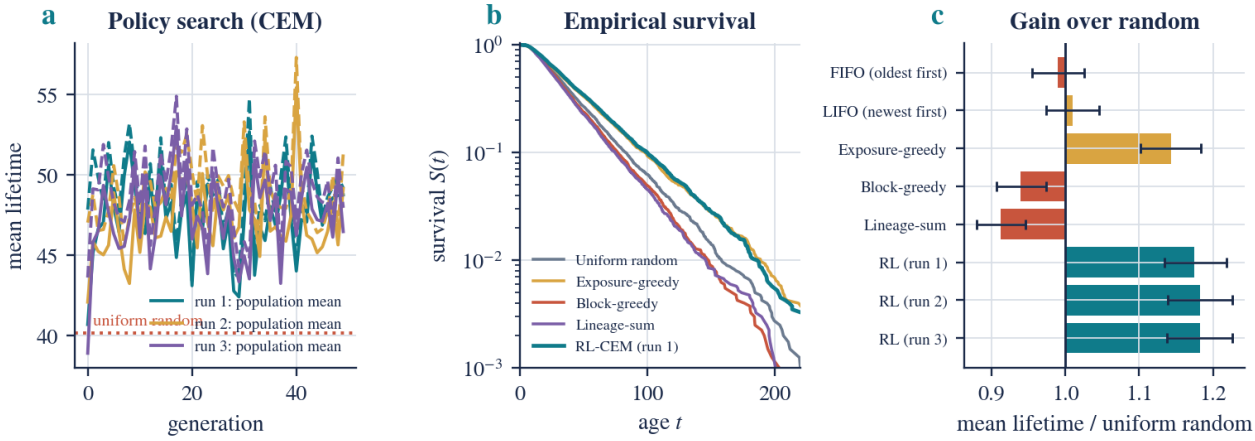


Figure 13. Policy search and its outcome. (a) Population mean (solid) and best candidate (dashed) over generations for three runs; the noise reflects fresh random seeds per generation. (b) Empirical survival on a log axis. (c) Mean-lifetime ratio to random order with bootstrap 95% intervals.

The learned weights (Figure 14a, normalised, since only the direction of the weight vector affects the ranking) share a pattern across runs: positive weight on the 8-unit block fraction and on exposure, negative weight on the 4-unit block fraction. The directions are only partly aligned (cosine similarities between 0.87 and 0.92), so the optimum is a broad ridge, not a point. We offer a hypothesis, which we did not test: serving units that threaten many intact neighbours cuts transmission, whereas serving the densest local clusters leaves isolated damaged units free to seed new clusters.

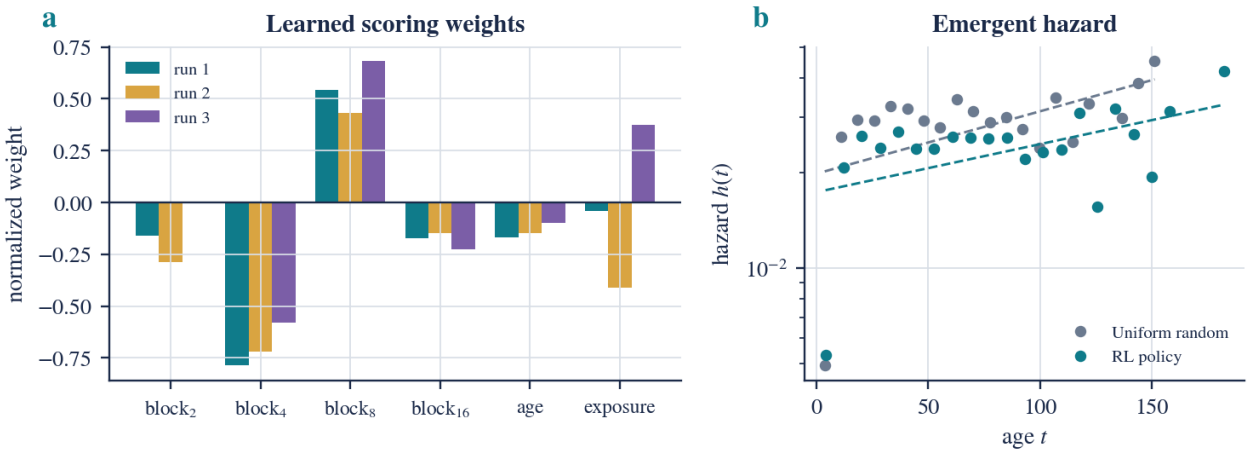


Figure 14. What was learned. (a) Normalised weights from three runs. (b) Empirical hazard against age with a log-linear (Gompertz) fit; the fit explains little variance.

Panel (b) and the straight survival lines of Figure 13b show that the emergent hazard is close to constant: a log-linear fit has $R^2 = 0.23$. This is expected from the theory, since first passage from a quasi-stationary state is nearly exponential. The model alone therefore does not reproduce Gompertz aging; Section 12 shows what does.

11.1 Robustness: contagion and budget

Table 12. Effect of contagion strength β ($B = 3$). The policy is retrained at each value; 1200 evaluation lives.

β	Random	RL (retrained)	RL / random (95% CI)	Block / random	FIFO / random
0.5	126.7	180.7	1.426 (1.33–1.54)	0.902	0.972
0.6	66.1	82.6	1.250 (1.16–1.35)	0.910	0.971
0.7	39.3	47.0	1.197 (1.12–1.29)	0.951	0.982
0.85	23.9	26.7	1.114 (1.05–1.18)	0.975	1.006
1	17.7	18.9	1.069 (1.01–1.13)	0.980	1.001
1.2	12.6	13.1	1.038 (0.98–1.10)	0.994	0.996

Table 13. Effect of the number of repair slots B (β = 0.7).

B	Random	RL (retrained)	RL / random (95% CI)	Block / random	FIFO / random
2	21.7	23.9	1.100 (1.04–1.16)	0.981	0.987
3	39.3	47.1	1.200 (1.12–1.28)	0.951	0.982
4	78.5	100.0	1.273 (1.18–1.37)	0.916	0.990
5	164.4	214.4	1.304 (1.21–1.40)	0.877	1.004
6	344.3	497.7	1.445 (1.34–1.56)	0.895	1.023

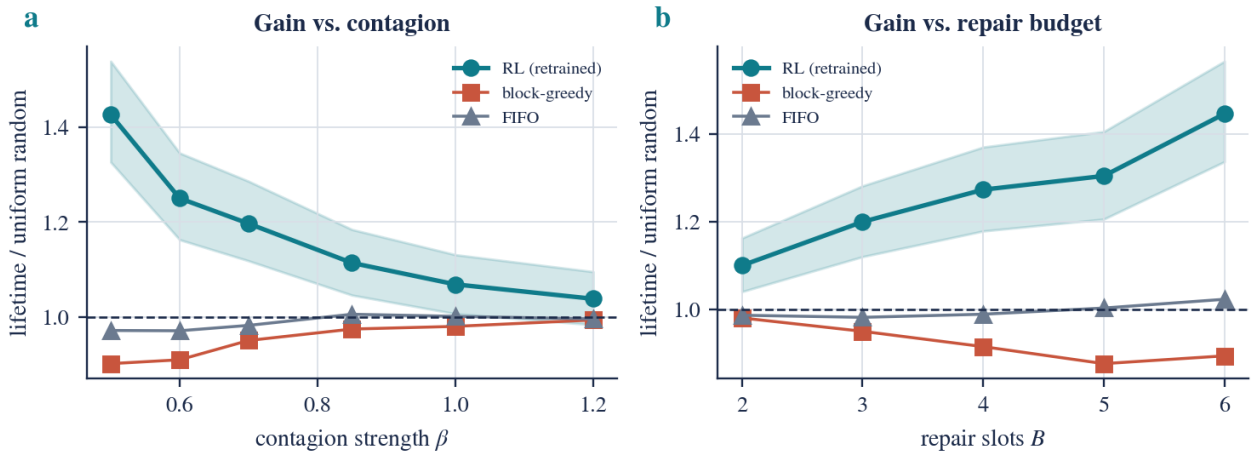


Figure 15. Gain from scheduling. (a) Against contagion strength; (b) against repair slots. Shaded bands are bootstrap 95% intervals for the learned policy.

Numerical result. The gain from learned ordering *decreases* with contagion, from +43% at $\beta = 0.5$ to +4% at $\beta = 1.2$, and *increases* with budget, from +10% at $B = 2$ to +44% at $B = 6$. Our initial hypothesis, that gains grow with damage correlation, is therefore falsified in this model. A reading consistent with the data is that ordering matters most when cascades are slow enough for repair to interrupt them yet the budget is tight enough that choices bind; strong contagion overwhelms any ordering, and weak budget leaves nothing to choose. This reading is an interpretation, not a tested mechanism.

12 The composite formula applied, and age-dependent capacity

12.1 Topology, efficiency and the exponent

We apply Conjecture 1 illustratively. For each graph of Section 5 we set $\theta_{\text{eff}} = \min(\varepsilon^*, p_{c,\text{meas}})$ with $\varepsilon^* = 0.429$, and evaluate Φ at $b = 0.3$ for the efficiencies found in Section 7 ($\lambda = 0.25$, $\mu = 1$). Because every measured p_c is below ε^*, the percolation threshold, not the decoding threshold, sets θ_{eff} in all four cases.

Table 14. Illustrative lifespan exponents from the composite formula (not an end-to-end simulation). $\Phi = 0$ means the system is in overload at that effective threshold.

Topology	κ	p_c (pred.)	p_c (meas.)	Φ, $\eta = 1$	Φ, $\eta = 0.886$	Φ, $\eta = 0.633$
Erdős–Rényi (c=4)	4.98	0.251	0.26	0.0105	0.0003	0.0105
Random 4-regular	4.00	0.333	0.34	0.0509	0.0068	0.0439
Barabási–Albert (m=2)	13.87	0.078	0.18	0.0000	0.0000	0.0000
Watts–Strogatz (k=4, p=0.1)	4.09	0.323	0.52	0.1107	0.0258	0.0929

Two qualitative lessons follow, both conditional on the conjecture. First, a network architecture that raises p_c (here the random-regular graph relative to the Barabási–Albert graph) changes the exponent more than any schedule at that threshold. Second, for a given architecture, a poor schedule can push a system from the protected regime into overload, as the $\eta = 0.633$ column shows for the Erdős–Rényi graph.

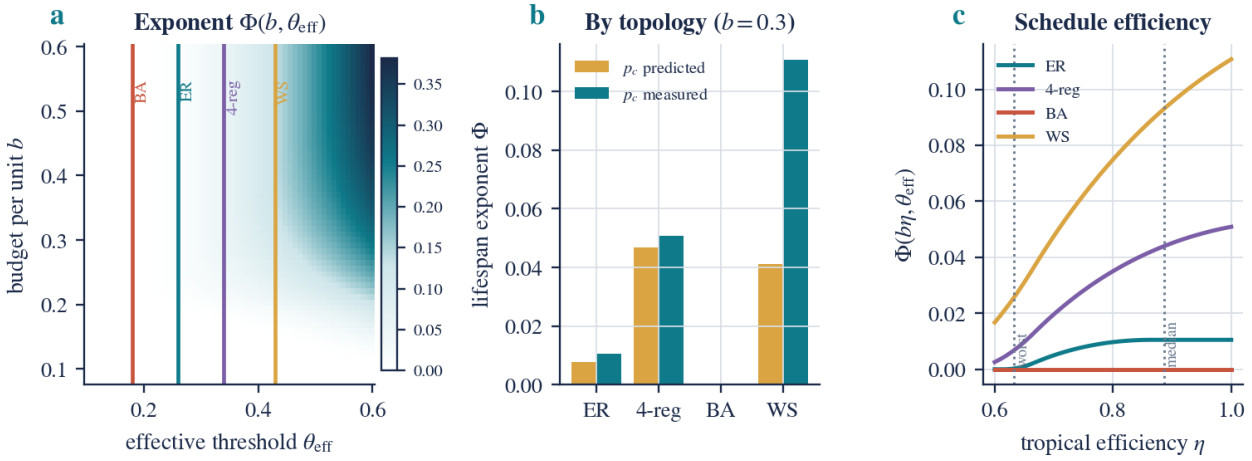


Figure 16. The composite formula in use. (a) Exponent over (θ_{eff}, b) with the four topologies marked. (b) Exponent by topology at $b = 0.3$ for predicted and measured p_c. (c) Exponent against tropical efficiency η.

12.2 Gompertz-like mortality from declining capacity

The constant hazard noted in Section 11 can be changed by one assumption: repair capacity declines with age, $b(t) = b_0 e^{-t/\tau}$. We solved the time-inhomogeneous Kolmogorov forward equations of the queue of Section 6 ($N = 60$, $b_0 = 0.45$, $\theta = 0.4$). Over the hazard range 10^{-4} to 0.5 the hazard is log-linear in age, as in Gompertz's law [13] (Table 15, Figure 17).

$$h(t) \approx h_0 e^{\gamma t}, \qquad \mathrm{MRDT} = \frac{\ln 2}{\gamma}, \qquad b(t) = b_0 e^{-t/\tau}. \tag{16}$$

Table 15. Gompertz fits to the hazard of the queue with exponentially declining capacity.

τ	Median lifespan	Gompertz rate γ	Mortality-rate doubling time	R² of log-linear fit
60	52.0	0.0826	8.4	0.978
120	88.5	0.0409	17.0	0.974
200	128.5	0.0243	28.5	0.963

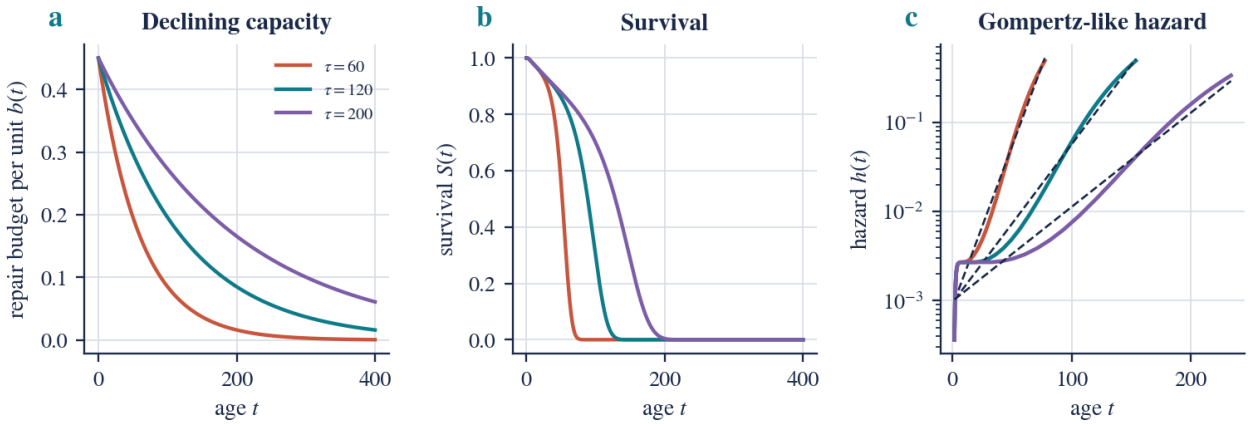


Figure 17. Declining capacity produces Gompertz-like hazard. (a) Budget per unit against age for three decline times τ. (b) Survival. (c) Hazard on a log axis with log-linear fits (dashed).

This is a demonstration of sufficiency, not evidence that capacity decline is the cause of Gompertz mortality, which many mechanisms could produce. It does connect the framework to a standard demographic observable: the doubling time of mortality scales with the decline time τ.

13 Predictions, status, and an experimental protocol

Table 16 lists the predictions of the framework with their current status, including those that failed.

Table 16. Predictions and their status.

#	Prediction	Status
H1	For exchangeable units, ordering at fixed budget has no effect on lifespan	Proved (Proposition 1)
H2	Scheduling gain grows with damage correlation (contagion)	**Falsified** in the hierarchical model (gain falls from +43% to +4%)
H3	Scheduling gain grows with repair budget	Supported in simulation (+10% to +44%)
H4	Extra budget gives no benefit beyond $b = \theta$	Proved and verified (Proposition 3, Figure 7)
H5	The best static split equalizes channel exponents	Proved (Proposition 5); verified numerically
H6	Ultrametric-aware greedy rules beat age-based rules	**Falsified** (−9% against −1% relative to random)
H7	Percolation threshold is $1/(\kappa-1)$	Supported for tree-like graphs; fails for scale-free and clustered graphs
H8	Composite threshold $\theta_{\text{eff}} = \min(\varepsilon^*, p_c)$	**Untested** end to end (Conjecture 1)
H9	Capacity decline yields Gompertz-like hazard	Supported in the model; assumption-dependent
H10	In a living system, reordering repair at matched flux changes lifespan	**Untested**; protocol below

13.1 A pre-registrable experiment

The decisive question is H10. We propose a lifespan assay in a short-lived model organism such as *C. elegans* [17], in which the order, but not the total throughput, of two repair pathways is changed by a tunable timing element; the exact genetic implementation is a design choice for the experimenters and is not specified here. Figure 18 gives the workflow. Three controls matter most: a matched-flux control, a knob-off arm, and, by Proposition 1, an arm whose damage is made as independent as possible, in which the predicted gain is zero.

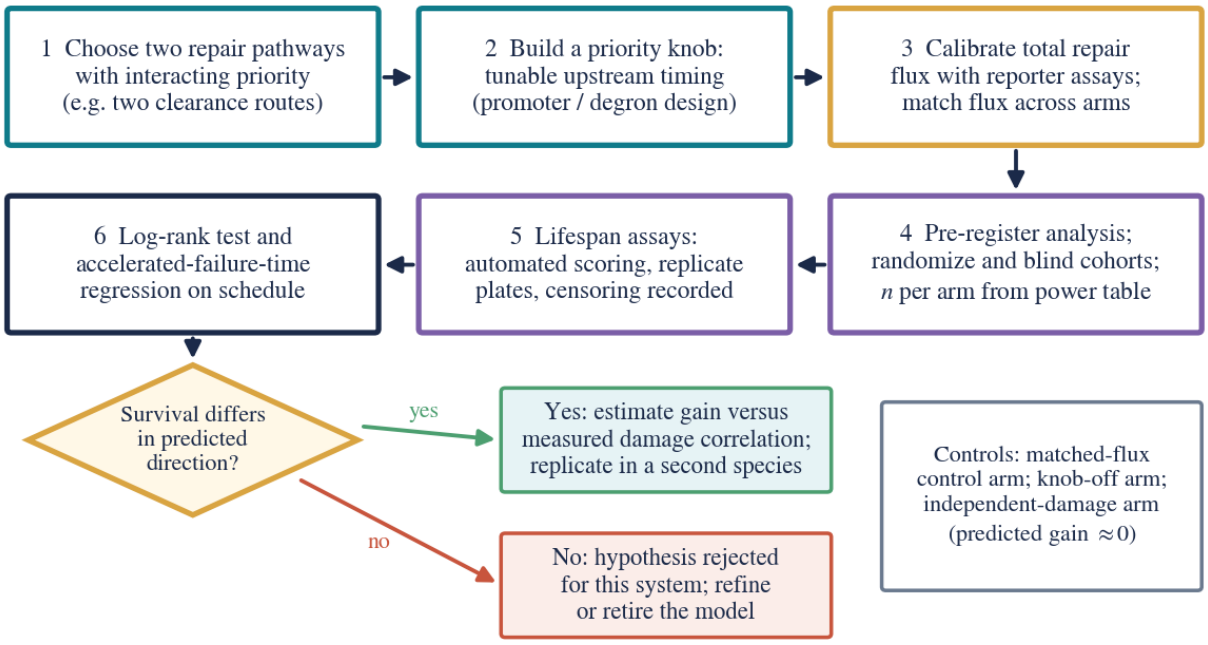


Figure 18. Proposed protocol. Steps 1 to 6 lead to a pre-specified decision on whether survival differs in the predicted direction at matched repair flux.

For two arms of equal size with full follow-up, the number of deaths needed for a log-rank test [49] at two-sided $\alpha = 0.05$ and power $1-\beta$ follows Schoenfeld's formula [48]:

$$d = \frac{4\,(z_{1-\alpha/2} + z_{1-\beta})^2}{(\ln \mathrm{HR})^2}, \qquad \mathrm{HR} = (1+g)^k, \tag{17}$$

where HR is the hazard ratio. The second relation holds if lifespans are Weibull with shape k and the manipulation stretches lifespan by a factor $1+g$. Figure 19 shows analytic power curves with Monte Carlo checks (for example, at $k = 5$ and $g = 6\%$, 60 animals per arm give simulated power 0.37 against 0.36 analytic), and the heat map lists animals per arm for 80% power. The Weibull shape is illustrative; it must be estimated from pilot survival curves.

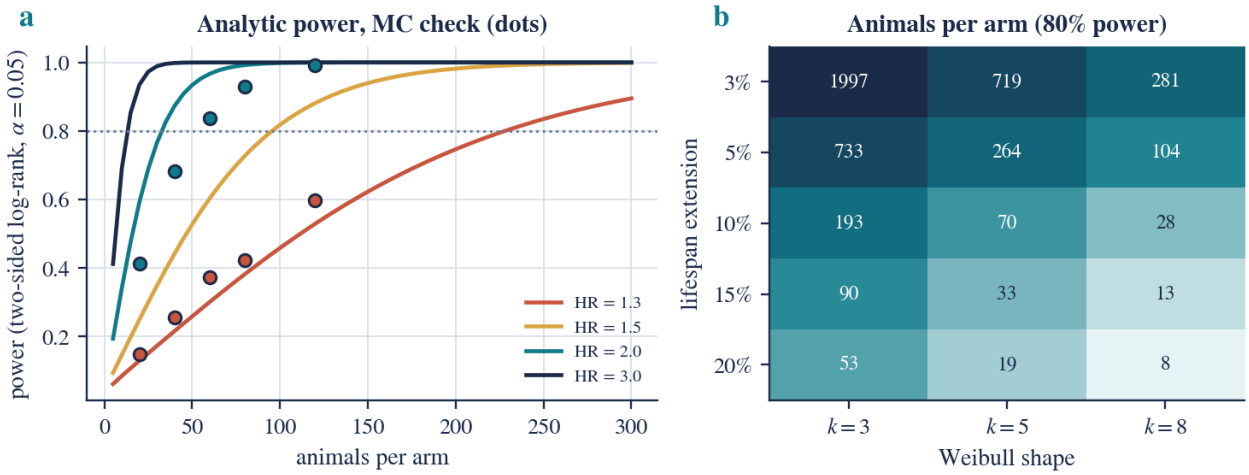


Figure 19. Sample size. (a) Power against animals per arm for four hazard ratios (lines, analytic) and Monte Carlo estimates (dots) for 6% and 12% lifespan extension at $k = 5$. (b) Animals per arm for 80% power by lifespan extension and Weibull shape.

14 Limitations

Abstraction. Units, rates, durations and thresholds are stylised and dimensionless. No parameter is fitted to biological data, so no numerical lifespan gain here should be read as a prediction for an organism. **Mapping.** The identification of repair with erasure decoding, of organismal failure with percolation or overload, and of tasks with a single shared machine are modelling assumptions. **Conjecture.** The composite formula is verified only component-wise. **Heuristic selection.** The learned policy is limited to a six-feature linear class; a richer class might do better, and the non-uniqueness of the learned weights shows the objective is flat. **Exponential lifetimes.** Without an additional assumption, the model gives near-constant hazard. **Percolation.** The tree-like formula fails for the scale-free and clustered graphs tested. **Search.** We did not perform an exhaustive literature search; parts of the composition may exist elsewhere. **Single instance.** Each stylised system is one parameter set; sweeps in β and B are the only robustness checks.

15 Discussion

The results sketch a consistent picture. Redundancy sets how much damage can be tolerated, and the network sets how much damage can accumulate before it spreads. Given those two numbers, budget buys exponential lifespan until it saturates, and how the budget is spent can multiply or divide the benefit. Scheduling is not a free lunch: in this study it is worth a few tens of percent in a hierarchical network and a factor of two or three in a tightly constrained cascade, and nothing at all when units are interchangeable. The most informative failures were the heuristics that made intuitive sense and lost to random choice, which is a reason to prefer computed policies over plausible ones.

For biology, the framework suggests a specific, testable direction: measure repair *flux*, *ordering* and *damage correlation* together, not only repair capacity. If a matched-flux reordering of two pathways shifts survival, the

effect size would calibrate the gain on a living system; if it does not, the model is wrong for that system, and that is an equally useful result. Translation to human lifespan is a distant question that this work does not address, and it would raise safety and ethical issues that experiments in model organisms must precede.

16 Conclusion

We posed the question of whether the order of repair matters at fixed capacity, and answered it within a model that joins six mathematical fields. Ordering is irrelevant for exchangeable units; with structure, the schedule changes throughput by up to 37% at equal work, extends the exact expected lifetime of a two-class cascade by 78% over the best static rule, and lets a learned policy outperform random order by 17–18% in a 64-unit hierarchy, while some plausible heuristics fail. Two of our own hypotheses were falsified and one central conjecture remains untested. The next step is an experiment, and Section 13 specifies one.

16.1 Acknowledgments and disclosures

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Appendix A: Proofs

Proof of Proposition 1. Under the stated assumptions the rate of damage events is $\tilde{\lambda}(D)$, a function of the count only, and each repair event removes one damaged unit at total rate $\mu \min(D,B)$, whichever units are served, because a work-conserving policy serves $\min(D,B)$ damaged units and every served unit is repaired at rate μ. Hence $(D(t))$ is a continuous-time Markov chain with policy-independent rates, and the failure set depends on D alone, so the law of T is policy-independent. ∎

Proof of Proposition 2. Fix level j and let f be a wavelet supported on a block of size p^j, constant on its p sub-blocks and summing to zero over them. Take x in the support. Targets y at distance $k < j$ lie in the same sub-block, so $f(y) = f(x)$ and contribute zero. Targets at distance j number $(p-1)p^{j-1}$; they lie in the other sub-blocks, whose values sum to $-f(x)$ per leaf-count p^{j-1}, giving $\sum(f(y)-f(x)) = -p^j f(x)$. Targets at distance $k > j$ lie outside the support, where $f = 0$, and there are $(p-1)p^{k-1}$ of them, contributing $-(p-1)p^{k-1} f(x)$. Summing with the rates w_k gives Eq. (13). For x outside the support, $\mathscr{L}f(x) = 0$ by the same cancellation. Counting wavelets gives the multiplicities, and together with the constant they span the space. ∎

Proof of Proposition 3. For a birth–death chain reflecting at 0, the expected time to pass from i to $i+1$ is $\tau_i = (u_i\pi_i)^{-1} \sum_{j\le i} \pi_j$ (flow balance across the cut between i and $i+1$), and $\mathrm{E}T = \sum_{i<m} \tau_i$. Write $\ln \pi_D = -\sum_{k\le D} \ln(d_k/u_{k-1}) \approx -N(V(x) - V(x^*))$ up to the normalisation, where $V'(x) = \ln[\mu \min(x,b)/(\lambda(1-x))]$ and x^* is its zero. The sum $\sum_{j\le i} \pi_j$ is of order one once i exceeds x^*N, so $\tau_i \approx \exp(N(V(i/N) - V(x^*)))/u_i$, and the sum over i is dominated by $i = m$, giving $\Phi = V(\theta) - V(x^*)$. If $x^* \ge \theta$ the drift is toward failure and the time is of order $\ln N$. ∎

Proof of Proposition 4. Each constraint $x_v(k) \ge x_u(k-t) + w$ is a max-plus inequality, and the minimal solution satisfies Eq. (9). By the standard theory of irreducible max-plus systems [30, 33], $x(k)$ grows asymptotically at rate equal to the maximum cycle mean of A, which equals the maximum ratio weight/tokens over circuits of the event graph; the formula in terms of powers is the standard trace characterisation. The circuit that visits every task through the sequencing arcs has weight $\sum d_i$ and exactly one token, so $\lambda(A) \ge \sum d_i$ and $\eta \le 1$. Equality holds when no other circuit has a larger ratio, i.e. when no precedence lag lies on a circuit that forces the machine to wait. ∎

Proof of Proposition 5. For independent channels $T = \min(T_1,T_2)$, the hazards add, and in the large-N limit the exponent of the mean is $\min(c_1,c_2)$ (Eq. 11). If c_1 is strictly increasing and c_2 strictly decreasing in α, then $\min(c_1,c_2)$ increases with α while $c_1 < c_2$ and decreases afterwards, so the maximum is at the unique crossing, which exists by the intermediate value theorem because the signs of $c_1 - c_2$ differ at the ends. ∎

Proof of Proposition 6. The binary erasure channel with erasure probability ε has capacity $1-\varepsilon$ [1]. A code of rate R admits vanishing error only if $R \le 1-\varepsilon$, so $\varepsilon \le 1-R$. A regular (d_v, d_c) ensemble has design rate $R = 1 - d_v/d_c$ or higher, so $\varepsilon^* \le d_v/d_c$. ∎

Appendix B: Algorithms and reproducibility

Peeling decoder. Build a random bipartite graph by randomly pairing $N\,d_v$ variable sockets with $N\,d_v$ check sockets; erase each variable with probability ε; repeat synchronous rounds in which every check with exactly one erased neighbour recovers it; stop when no recovery occurs. Trials: 300, 150, 60 for $N = 500, 2000, 8000$.

```
Gillespie simulation of one lifetime (hierarchical model)
  state s[64], field f[i] = sum_j W[i,j] s[j], counts per block at levels 1..4
  loop:
    serve = top-B damaged units by score w . phi(i)   (random scores if policy = random)
    rates: intact i -> lam + beta*f[i];  served i -> mu
    t += Exp(total rate); pick event by rate; update s, f, block counts
    if any 8-block count >= 7 or total damaged >= 32: return t

Cross-entropy method
  mean = 0, std = 1; repeat 50 generations:
    sample 40 weight vectors ~ N(mean, std^2)
    evaluate each on 300 lives (common seeds within a generation)
    refit mean, std to the 8 best (std += 0.03)
```

Exact dynamic programming. States (d_1, d_2) with $d_1 < 9$, $d_2 < 20$; shares α on a 21-point grid; policy iteration (solve the sparse linear system for the expected absorption time, then improve the policy) until the policy is stable. Survival functions use the matrix exponential of the generator on a grid of step 6 time units.

Seeds and software. Python 3.12 with NumPy, SciPy, NetworkX, Numba and Matplotlib; random generators were seeded as described in the code (training seeds and evaluation seeds are disjoint). Percolation used five replicates per density; the Weibull power check used 1500 simulated experiments per point. The tropical enumeration is deterministic.